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Two electrodes E2/E4 -> four-resistor-per-leg input networks -> dual AMC3330 isolated paths -> independent ADS1115 converters -> ESP32.

HI: 997:1 divider; 600 VAC RMS design target. LO: 10.96:1 divider; use conservative +/-2 V working range before characterization.

Both ADCs sample continuously at 860 SPS nominal; separate unsynchronized clocks. DIAG and READY are wired to ESP32.

2S harness -> F1 -> Q1 -> protected VBAT. Motor/ESC power stays external. Never parallel ESC BEC.

Remove JP1 before USB. U1 requires WROOM pin availability. Expansion signals are 3.3 V only.

No CAT rating, surge qualification, water-safety threshold or guaranteed detection floor is claimed.

Control

Power

Input_protection

Isolation

Health

ADC

File: Control.kicad_sch

File: Power.kicad_sch

File: Input_protection.kicad_sch

File: Isolation.kicad_sch

File: Health.kicad_sch

File: ADC.kicad_sch

Engineering prototype – qualification required

IoT Beyond Lab

Sheet: /

File: EWS_RevH.kicad_sch

Title: EWS Rev H / System overview

Size: A3Date:KiCad E.D.A. 10.0.6

Rev: H prototypeId: 1/7

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